BOOSTING SCORES FOR NEARBY UNIVERSITIES USING GEOLOCATIONS

Let's say we are searching for universities in the United Kingdom. While searching, we want to give preference to universities within a 10 km radius of London Bridge. We boost the score for them.

To try this scenario, let's create a mapping for the `university` index with a location declared as a `geo_point` field. The following listing creates the mapping and the indexes for four universities: two in London and two elsewhere in the country.

Listing 12.20 Creating a `universities` index

```
PUT universities
{
  "mappings": {
    "properties": {
      "name":{
        "type": "text"
      },
      "location":{
        "type": "geo_point"
      }
    }
  }
}

PUT universities/_doc/1
{
  "name":"London School of Economics (LSE)",
  "location":[0.1165, 51.5144]
}

PUT universities/_doc/2
{
  "name":"Imperial College London",
  "location":[0.1749, 51.4988]
}

PUT universities/_doc/3
{
  "name":"University of Oxford",
  "location":[1.2544, 51.7548]
}

PUT universities/_doc/4
{
  "name":"University of Cambridge",
  "location":[0.1132, 52.2054]
}
```

Now that the index and data are prepped, let's fetch universities, boosting the relevance scores so those closer to London Bridge are at the top of the list. The map of London in figure 12.12 shows the approximate distances of these universities from London Bridge and Knightsbridge. We use the `distance_feature` query for this